

NURUL HASAN

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PROFESSIONAL SUMMARY

B.Tech CSE (AI & ML) student with hands-on experience developing backend applications using Java and Spring Boot. Skilled in REST API development, Spring Security, Hibernate/JPA, PostgreSQL, and MySQL, with practical exposure to microservices architecture, API Gateway, Eureka Service Discovery, Apache Kafka, Event-Driven Architecture, Docker, and Git/GitHub. Currently developing a microservices-based chatting application and has completed backend projects through a CodeAlpha internship and a hackathon.

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL

Backend Development: Spring Boot, Spring Web, Spring WebFlux, FastAPI, REST APIs

Java & Persistence: Spring Security, Hibernate, JPA, Spring Data JPA, Maven

Microservices & Distributed Systems: Microservices Architecture, API Gateway, Eureka Service Discovery, Apache Kafka, Event-Driven Architecture

Databases: PostgreSQL, MySQL

Security: JWT, Authentication, Authorization, HttpOnly Cookies

DevOps & Tools: Docker, Linux, Git, GitHub, IntelliJ IDEA, Postman

AI/ML: Machine Learning, Natural Language Processing (NLP), RAG Fundamentals

EXPERIENCE

CodeAlpha — Backend Development Intern

Completed Internship

- Engineered backend application components using Java, Spring Boot, and REST API standards as part of the CodeAlpha internship program.
- Implemented secure authentication and authorization mechanisms using Spring Security and JWT.
- Configured database persistence layers using JPA/Hibernate for MySQL and PostgreSQL database integration.
- Managed project build lifecycles with Maven, containerized application components using Docker, and maintained version control via Git and GitHub.

PROJECTS

Microservices Architecture Chatting Application

Currently In Progress

Technologies: Java, Spring Boot, Spring Security, Kafka, Eureka Server, API Gateway, Hibernate, JPA, PostgreSQL, REST APIs, Spring WebFlux

- Designing a scalable microservices architecture by building independent backend services using Spring Boot and REST APIs.
- Configured an API Gateway for centralized routing, security enforcement, and JWT-based authentication and authorization with HttpOnly cookies.
- Utilized Eureka Server for dynamic service discovery and Apache Kafka to enable asynchronous, event-driven service-to-service communication.
- Developed Authentication and User backend services backed by PostgreSQL with Hibernate/JPA, while exploring reactive backend development using Spring WebFlux.
- Architected system design with Redis planned as a future caching implementation for user presence and performance optimization.

Savory Restaurant Management System — CodeAlpha Internship Project

Technologies: Java, Spring Boot, MySQL, Spring Security, JPA, JWT, REST APIs, Docker

- Developed a backend-driven management system utilizing Java, Spring Boot, and REST APIs.
- Implemented Spring Security with JWT to establish secure authentication and role-based authorization.
- Designed database schemas using MySQL with JPA/Hibernate for object-relational mapping and containerized the service using Docker.

URL Shortener — CodeAlpha Internship Project

Technologies: Java, Spring Boot, PostgreSQL, Spring Data JPA, Spring Security, REST APIs, Docker

- Built a URL-shortening backend application using Spring Boot, delivering REST APIs for short-code generation, URL validation, and redirection.
- Integrated PostgreSQL using Spring Data JPA for persistent link data management and validation logic.
- Configured Spring Security and packaged application artifacts with Docker and Maven for reliable local environment execution.

Fraud/Phishing Email Detector — Hackathon Project

Technologies: Python, Machine Learning, NLP

- Developed an AI/ML-based email classification system to detect fraudulent and phishing content during a hackathon.
- Applied Natural Language Processing (NLP) techniques for text preprocessing, feature extraction, and training machine learning classification models.

EDUCATION

Brainware University — Barasat, West Bengal

2024 – 2028

B.Tech in Computer Science & Engineering (Artificial Intelligence & Machine Learning)

LANGUAGES

Bengali: Fluent | **Hindi:** Fluent | **English:** Working Proficiency